ISOENERGETIC KAM, CONFINEMENT AND THE RESTRICTED THREE-BODY PROBLEM: BLUEPRINT

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Abstract. A proof plan for the four frozen statements of track I: the isoenergetic a-posteriori KAM theorem (I-G1), all-time action confinement for two degrees of freedom (I-HEAD), the Kepler bordered determinant (I-KEPLER), and the near-circular region of the restricted circular planar three-body problem (I-R3BP). Results of the blueprint of track K are cited by their labels [K:`bp:p1:...`] and [K:`bp:p2:...`]. Every result is labelled and cites its pinned source.

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1. Conventions, definitions and the frozen statements

This part fixes notation and restates, in mathematical form, the four frozen theorems of track I and the frozen definitions they use: the fifteen definitions of `KAM/Frozen/Iso/Defs/` (owner approval D20, 27 October 2026) and the definitions of `KAM/Frozen/Defs/` of track K. The Lean declarations are authoritative; the statements below are faithful transcriptions, with the quantifier order and the order in which constants are chosen exactly as in Lean. Results of track K's blueprint are cited as [K:`bp:p1:...`] and [K:`bp:p2:...`]; they are not re-proved here.

1.1. Conventions.

- Ambient spaces are $\mathbb{R}^m = (\texttt{Fin}\ m \to \mathbb{R})$ and $\mathbb{C}^m = (\texttt{Fin}\ m \to \mathbb{C})$. Angles are $\theta \in \mathbb{C}^n$, actions $I \in \mathbb{C}^n$, phase points $z = (\theta, I) \in \mathbb{C}^n \times \mathbb{C}^n$. Functions of θ are 1-periodic, as in Part 0 of track K ([K:`bp:p0:conventions`]).
- *Norms.* $|v| = \|v\| = \max_i |v_i|$ on $\mathbb{R}^m$ and $\mathbb{C}^m$; on a product the maximum of the two component norms. This is the norm of `Metric.ball`, of `dist` in `actionNbhd`, of $\|DH\|$ (operator norm of a $\mathbb{C}$-linear map, `fderiv`), and of $\|D^2H\|$ (norm of the bilinear map `iteratedFDeriv`). A row-sum bound $\sum_j |M_{ij}| \le c$ for every i is written $|M| \le c$. The norm of a complex number is its modulus.
- The strip $T_\rho = \{\theta \in \mathbb{C}^n : |\operatorname{Im}\theta_i| < \rho\ \forall i\}$ is open; it contains $\mathbb{R}^n$ and is invariant under $\mathbb{Z}^n$ and $\mathbb{R}^n$ translations. The set `Set.pi univ (Icc 0 1)` $= [0,1]^n$ carries Lebesgue measure, of total mass one; $\langle u \rangle = \int_{[0,1]^n} u(\theta)\, d\theta$ for a function evaluated at real angles. Averages of vectors or matrices are entrywise.
- A map is *holomorphic on an open set U* if it is `DifferentiableOn` $\mathbb{C}$ there. On an open set all derivatives `fderiv`, `iteratedFDeriv` used below are genuine (Lean totality, Remark 1.13).

- Real and complex points. A real vector $x \in \mathbb{R}^m$ is identified with $(x_i) \in \mathbb{C}^m$; "$(I : \mathbb{C}) \in S$" for $S \subset \mathbb{C}^n$ means this identification lies in S. The real phase points of $D \subset \mathbb{C}^n \times \mathbb{C}^n$ are the $(\theta, I) \in \mathbb{R}^n \times \mathbb{R}^n$ whose image lies in D.
- *Two phase-space conventions.* In I-G1 and I-HEAD, a phase point is (θ, I) (angles first) and $X_H = (\partial_I H, -\partial_\theta H)$. In the restricted three-body problem a phase point is $z = (q, p)$ with position $q \in \mathbb{R}^2$ and canonical momentum $p \in \mathbb{R}^2$ (position first, as in the Lean carrier $(\texttt{Fin}\ 2 \to \mathbb{R}) \times (\texttt{Fin}\ 2 \to \mathbb{R})$); there `radius` is the Euclidean length, not the max norm. The norm $\|\cdot\|$ of I-HEAD is the max norm; the bounds in the conclusion of I-R3BP use rad.

1.2. **Frozen definitions of track K used by track I.**

Definition 1.1 (Strip, action neighbourhood)**.** $T_\rho = \{\theta \in \mathbb{C}^n : \forall i,\ |\operatorname{Im}\theta_i| < \rho\}$ (`complexStrip` n ρ). For $B \subset \mathbb{R}^n$ and $r \in \mathbb{R}$, $B_r = \{I \in \mathbb{C}^n : \exists J \in B,\ \|I - J\| < r\}$ (`actionNbhd`); it is open. Thus for real I: $I \in B_r$ iff $\exists J \in B$ with $\|I - J\| < r$.
Frozen Lean declaration: `KAM.complexStrip, KAM.actionNbhd`
Source: HL19 §2.1; see [K:`bp:p0:domains`].

Definition 1.2 (Diophantine vectors)**.** $\omega \in \mathbb{R}^n$ is (γ, τ)-Diophantine if for every $k \in \mathbb{Z}^n \setminus \{0\}$ $\gamma / \big(\sum_i |k_i|\big)^\tau \le |k \cdot \omega|$ (real power).
Frozen Lean declaration: `KAM.IsDiophantine`
Source: HL19 (3.1), p. 14; see [K:`bp:p0:diophantine`].

Definition 1.3 (Hamiltonian vector fields)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$ and a real phase point z: $X_H(z) = \big(i \mapsto DH(z)(0, e_i),\ i \mapsto -DH(z)(e_i, 0)\big)$, with D the real Fréchet derivative. For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ the same formula with the complex derivative gives $X_H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$. Thus $X_H = (\partial_I H, -\partial_\theta H)$.
Frozen Lean declaration: `KAM.hamiltonianVectorField, KAM.hamiltonianVectorField`$\mathbb{C}$
Source: HL19 Rem. 2.1, p. 6.

Definition 1.4 (Real Hamiltonian, frequency map)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and $f : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$ and a real phase point (θ, I), $\mathcal{H}_{h,f}(\theta, I) = \operatorname{Re}\big(h(I) + f(\theta, I)\big)$ (`realHamiltonian`). The frequency map is $\nabla h(I)_i = D\big(J \mapsto \operatorname{Re} h(J)\big)(I)\, e_i$ for real I (`frequencyMap`), a real function $\mathbb{R}^n \to \mathbb{R}^n$.
Frozen Lean declaration: `KAM.realHamiltonian, KAM.frequencyMap`
Source: see [K:`bp:p0:nearlyintegrable`].

Definition 1.5 (Torus data on strips)**.** Let $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$.

- K is *primary on* T_ρ if $K(\theta + k) = K(\theta) + (k, 0)$ for $\theta \in T_\rho$, $k \in \mathbb{Z}^n$ (`IsPrimaryOnStrip`); *real on reals* if every $K(\theta)$, $\theta \in \mathbb{R}^n$, is a real phase point (`IsRealOnReals`).
- Bilinear dot product $u \cdot v = \sum_i (u_{1,i} v_{1,i} + u_{2,i} v_{2,i})$ on $\mathbb{C}^n \times \mathbb{C}^n$ (`dot`$\mathbb{C}$), $\Omega_0(a, b) = (-b, a)$ (`omegaZero`).
- Invariance error $E_{H,K,\omega}(\theta) = X_H(K(\theta)) - DK(\theta)\omega$ (`invarianceError`).
- Gram matrix $G_K(\theta)_{ij} = DK(\theta)e_i \cdot DK(\theta)e_j$ (`gramMatrix`), a complex $n \times n$ matrix.
- Normal frame $N_K(\theta)_{\cdot j} = \sum_k (G_K^{-1})_{kj}\, \Omega_0\big(DK(\theta)e_k\big) \in \mathbb{C}^n \times \mathbb{C}^n$ (`normalFrame`), with the matrix inverse of Lean.
- Torsion $T_{H,K,\omega}(\theta)_{ij} = N_{\cdot i}(\theta) \cdot \Omega_0\big(DX_H(K(\theta)) N_{\cdot j}(\theta) - D_\theta(N_{\cdot j})(\theta)\,\omega\big)$ (`torsion`); $N_{\cdot j}$ is regarded as a function of θ.

Frozen Lean declaration: `KAM.IsPrimaryOnStrip, KAM.IsRealOnReals, KAM.dot`$\mathbb{C}$`, KAM.omegaZero, KAM.invarianceError, KAM.gramMatrix, KAM.normalFrame, KAM.torsion`
Source: HL19 (2.2), (3.2)–(3.9), pp. 7, 16–17; see [K:`bp:p0:stripdata`].

1.3. **Frozen definitions of track I.**

Definition 1.6 (Energy average and extended torsion)**.** For $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, K as above, $\omega \in \mathbb{R}^n$, $\theta \in \mathbb{C}^n$, with index set $\{1, \dots, n\} \sqcup \{*\}$:

$$\bar{E}_H(K) = \int_{[0,1]^n} H\big(K(\theta)\big)\, d\theta \in \mathbb{C}, \qquad \mathsf{T}_H(\theta) = \begin{pmatrix} T_{H,K,\omega}(\theta) & \omega \\ \big(DH(K(\theta))N_{\cdot b}(\theta)\big)_b & 0 \end{pmatrix},$$

the $(n+1) \times (n+1)$ complex matrix whose upper-left block is the torsion, last column $(\omega, 0)$ (real entries, viewed in $\mathbb{C}$), and last row $b \mapsto DH(K(\theta))\big(N_{\cdot b}(\theta)\big)$ (complex derivative of H applied to the normal frame), last entry 0. The averaged extended torsion is $\langle \mathsf{T}_H \rangle = \big(\int_{[0,1]^n} \mathsf{T}_H(\theta)_{ij}\, d\theta\big)_{ij}$, the integral taken over real angles.

Frozen Lean declaration: `KAM.Iso.energyAverage`, `KAM.Iso.extendedTorsion`, `KAM.Iso.extendedTorsionAverage`

Source: HL19 (3.13)–(3.14), p. 19, with $c = H$, $d = n$ (Case III), $\hat{\omega} = \omega$; the last row is $DH(K)N$ even off invariance, not $\omega^\top$ (HL19 (3.17) is the invariant case, Rem. 3.7, p. 19).

Definition 1.7 (Bordered Hessian)**.** For $h : \mathbb{C}^n \to \mathbb{C}$ and real I, the real $(n+1) \times (n+1)$ matrix

$$\mathsf{B}_h(I) = \begin{pmatrix} \big(\partial_b(\nabla h)_a(I)\big)_{a,b} & \nabla h(I) \\ \nabla h(I)^\top & 0 \end{pmatrix},$$

where $\partial_b(\nabla h)_a(I) = D(\nabla h)(I)e_b$ read at a, the real derivative of the frequency map (`borderedHessian`). No symmetry is assumed in the definition.

Frozen Lean declaration: `KAM.Iso.borderedHessian`

Source: Chi05 p. 17 (display after (3.43)); HL19 Rem. 3.7, p. 19.

Definition 1.8 (Hamiltonian solutions with action in B)**.** For $H : \mathbb{R}^n \times \mathbb{R}^n \to \mathbb{R}$, $B \subset \mathbb{R}^n$, $J \subset \mathbb{R}$ and $z : \mathbb{R} \to \mathbb{R}^n \times \mathbb{R}^n$, write $\mathrm{Sol}(H, B, J, z)$ if for all $t \in J$: $z(t)_2 \in B$, H is real differentiable at $z(t)$, and z is differentiable at t with $z'(t) = X_H(z(t))$ (`HasDerivAt`).

Frozen Lean declaration: `KAM.Iso.IsHamiltonianSolutionOn`

Source: ours (D20); the domain B and the differentiability of H are part of the solution concept.

Definition 1.9 (Planar Euclidean length, the primaries, collisions)**.** For $q \in \mathbb{R}^2$: $\rho_2(q)^2 = q_0^2 + q_1^2$ (`radiusSq`) and $\mathrm{rad}(q) = \sqrt{q_0^2 + q_1^2}$ (`radius`; the radicand is nonnegative). For $\mu \in \mathbb{R}$: $a_1(\mu) = (-\mu, 0)$, $a_2(\mu) = (1 - \mu, 0)$ ($\texttt{primary}_1$, $\texttt{primary}_2$). A phase point $z = (q, p) \in \mathbb{R}^2 \times \mathbb{R}^2$ is *collision free* (`CollisionFree` μ) if $\mathrm{rad}(q - a_1(\mu)) > 0$ and $\mathrm{rad}(q - a_2(\mu)) > 0$.

Frozen Lean declaration: `KAM.Iso.radiusSq`, `KAM.Iso.radius`, $\texttt{KAM.Iso.primary}_1$, $\texttt{KAM.Iso.primary}_2$, `KAM.Iso.CollisionFree`

Source: Chi05 (3.34), (3.37), p. 12–13 (barycentric rotating frame, unit separation, total mass one); ours.

Definition 1.10 (Restricted Hamiltonian and field)**.** On collision-free points $z = (q, p)$, with $r_j = \mathrm{rad}(q - a_j(\mu))$:

$$\mathcal{H}_\mu^{\mathrm{R3}}(q,p) = \tfrac{1}{2}|p|_2^2 + q_1 p_0 - q_0 p_1 - \frac{1-\mu}{r_1} - \frac{\mu}{r_2}, \qquad |p|_2^2 := p_0^2 + p_1^2,$$

$$V_\mu(q,p) = \Big((p_0 + q_1,\ p_1 - q_0),\ (p_1, -p_0) - \frac{1-\mu}{r_1^3}(q - a_1) - \frac{\mu}{r_2^3}(q - a_2)\Big).$$

Thus $V_\mu = (\partial_p \mathcal{H}_\mu^{\mathrm{R3}}, -\partial_q \mathcal{H}_\mu^{\mathrm{R3}})$ on the open collision-free set. A function $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ is a *restricted solution on* J, $\mathrm{RSol}_\mu(J, z)$, if for every $t \in J$ the point $z(t)$ is collision free and z is differentiable at t with $z'(t) = V_\mu(z(t))$.

Frozen Lean declaration: `KAM.Iso.restrictedHamiltonian`, `KAM.Iso.restrictedVectorField`, `KAM.Iso.IsRestrictedSolutionOn`

Source: Chi05 (1.2), p. 2 and (3.34)–(3.39), pp. 12–14, in barycentric rotating coordinates with canonical momentum (not rotating velocity); the equations V_μ are the Hamilton equations of $\mathcal{H}_\mu^{\mathrm{R3}}$ (the check is part of the proofs, see PLAN.md).

Definition 1.11 (Near-circular region)**.** For $a_-, a_+, e_-, e_+ \in \mathbb{R}$, $\mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ is the set of (q, p) for which there exist $r, a, e \in \mathbb{R}$ with

$$r > 0,\ r^2 = q_0^2 + q_1^2, \quad a > 0, \quad 0 < e < 1, \quad a_- < a < a_+, \quad e_- < e < e_+,$$

$$\tfrac{1}{2}|p|_2^2 - \frac{1}{r} = -\frac{1}{2a}, \qquad q_0 p_1 - q_1 p_0 > 0, \qquad (q_0 p_1 - q_1 p_0)^2 = a(1 - e^2).$$

(`NearCircular`.) The three numbers r, a, e are determined by (q, p), so no choice is involved.
Frozen Lean declaration: `KAM.Iso.NearCircular`
Source: Chi05 (2.24)–(2.26), p. 9: $E = -1/(2a)$, $\Gamma^2 = a(1 - e^2)$ in the units $\mu = M = 1$; referred to $\mu = 0$ (primary at the origin).

Definition 1.12 (The Kepler bordered matrix)**.** For $\kappa, \Omega \in \mathbb{R}$ and $L > 0$,

$$\mathsf{K}_{\kappa,\Omega}(L) = \begin{pmatrix} -3\kappa/L^4 & 0 & \kappa/L^3 \\ 0 & 0 & -\Omega \\ \kappa/L^3 & -\Omega & 0 \end{pmatrix} \in \mathbb{R}^{3\times 3},$$

the bordered Hessian $\left(\begin{smallmatrix} D^2 h & \nabla h \\ \nabla h^\top & 0 \end{smallmatrix}\right)$ of $h(L, G) = -\kappa/(2L^2) - \Omega G$ (`keplerBorder`, L in the subtype $\{L : \mathbb{R} \mid L > 0\}$).
Frozen Lean declaration: `KAM.Iso.keplerBorder`
Source: Chi05 p. 17 ($\kappa = \Omega = 1$).

1.4. **Lean totality.**

Remark 1.13 (Lean totality). The frozen definitions use `fderiv` (zero where a function is not differentiable), the matrix inverse (zero for a singular matrix), the Bochner integral (zero for a non-integrable function), division, `Real.sqrt` (zero for a negative argument) and real powers. The statements guard them as follows; each proof must establish the guarded facts before using the defining formulas.

- (T1) *Derivatives of H in I-G1.* D is open, H is holomorphic on D, so DH, X_H, DX_H and D^2H are genuine derivatives at every point of D; K is holomorphic on the open strip T_ρ and $B_\mu(K(\theta)) \subset D$, so $H \circ K$, $X_H \circ K$ and DK are genuine on T_ρ.
- (T2) *Inverses in I-G1.* The Gram matrix inverse is used only under the hypothesis $\det G_K(\theta) \in \mathbb{C}^\times$ on T_ρ, and the inverse of the averaged extended torsion only under $\det\langle \mathsf{T}_H \rangle \in \mathbb{C}^\times$. Then N_K is genuine and holomorphic on T_ρ (by the holomorphic inverse, [K:`bp:p1:holomorphic-inverse`]), so $DN_{\cdot j}$ and T are genuine. For the conclusion, invertibility of the Gram determinant for K_∞ must be proved (it is not a hypothesis on K_∞) before $E_{H,K_\infty,s_\infty\omega^*}$ is used; but note that $E = 0$ and $H \circ K_\infty = E_0$ use only DK_∞ and X_H, not the frame.
- (T3) *Integrals in I-G1.* The cube $[0, 1]^n \subset \mathbb{R}^n \subset T_\rho$ is compact, $\rho > 0$, and the integrands $H \circ K$ and the entries of T_H are continuous on it by (T1)–(T2), hence integrable ([K:`bp:p1:ext-integral`]). Integrability is not a hypothesis; it must be proved.
- (T4) *Powers and division in I-G1.* $\gamma > 0$ and $\rho > 0$, so γ^4, γ^2, $\rho^{4\tau}$, $\rho^{2\tau}$ are positive and the quotients are genuine; $\gamma/(\sum |k_i|)^\tau$ has a positive base since $k \neq 0$.
- (T5) *I-HEAD.* h is holomorphic on the open set B_{r_0}, so ∇h and $D(\nabla h)$ are genuine at real points of B_{r_0} (an open subset of $\mathbb{R}^2$) and B_h is genuine. The solution predicate Sol includes differentiability of $\mathcal{H}_{h,f}$ at each visited point, which also guards $X_{\mathcal{H}}$; it holds for $z(t)_2 \in B \subset B_{r_0}$ and real θ because f is holomorphic on $T_{s_0} \times B_{r_0}$.
- (T6) *I-KEPLER, I-R3BP.* $L > 0$ guards every denominator of $\mathsf{K}_{\kappa,\Omega}(L)$. Every inverse distance in $\mathcal{H}_\mu^{\mathrm{R3}}$, V_μ is guarded by collision freedom, and the radicand of `radius` is a sum of squares. In $\mathcal{N}$, $r > 0$ and $a > 0$ precede the energy equation.

1.5. **The frozen theorems.**

Theorem 1.14 (I-G1: isoenergetic a-posteriori KAM)**.** *Let $n \geq 2$ and $\tau \in \mathbb{R}$ with $\tau \geq n-1$; let $\rho, \rho_\infty \in \mathbb{R}$ with $0 < \rho_\infty < \rho$; let $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma}, \sigma_\omega, \nu \in \mathbb{R}$ with $M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma} > 0$, $\sigma_\omega > 1$, $\nu > 0$; let $\omega^* \in \mathbb{R}^n$, $\omega^* \neq 0$. Then there exist $C_1, C_2, C_3 > 0$ such that for all $D \subset \mathbb{C}^n \times \mathbb{C}^n$, $H : \mathbb{C}^n \times \mathbb{C}^n \to \mathbb{C}$, $K : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$, and $s, \gamma, e, E_0 \in \mathbb{R}$ satisfying*

- *(I1) D is open and $(\theta + k, I) \in D$ whenever $(\theta, I) \in D$, $k \in \mathbb{Z}^n$;*
- *(I2) H is holomorphic on D, and $H(\theta + k, I) = H(\theta, I)$ on D for $k \in \mathbb{Z}^n$;*
- *(I3) for every real phase point (θ, I) lying in D, $\operatorname{Im} H(\theta, I) = 0$;*
- *(I4) for every $z \in D$: $\|DH(z)\| \leq M$, $\|X_H(z)\| \leq M$, $\|D(X_H)(z)\| \leq M$, $\|D^2(X_H)(z)\| \leq M$, $\|D^2 H(z)\| \leq M$;*
- *(I5) K is holomorphic on T_ρ, primary on T_ρ, and real on reals;*
- *(I6) for every $\theta \in T_\rho$: $B_\mu(K(\theta)) \subset D$ (open max-norm ball) and $\|DK(\theta)\| \leq \sigma_K - \mu$;*
- *(I7) for every $\theta \in T_\rho$: $\det G_K(\theta) \neq 0$ and $|G_K(\theta)^{-1}| \leq \sigma_B - \mu$ (maximal row sum);*
- *(I8) $\det\langle \mathsf{T}_H \rangle \neq 0$ with T_H built from $(H, K, s\omega^*)$, and $|\langle \mathsf{T}_H \rangle^{-1}| \leq \sigma_T - \mu$ (maximal row sum);*
- *(I9) $1 < s < \sigma_\omega$, $\nu \leq (s-1)\|\omega^*\|$, $\nu \leq (\sigma_\omega - s)\|\omega^*\|$;*
- *(I10) $0 < \gamma \leq \bar{\gamma}$, $\omega^* \in \mathcal{D}_{\gamma,\tau}$, $0 \leq e$;*
- *(I11) $\|E_{H,K,s\omega^*}(\theta)\| \leq e$ for every $\theta \in T_\rho$;*
- *(I12) $|\bar{E}_H(K) - E_0| \leq e$;*
- *(I13) $C_1 e/(\gamma^4 \rho^{4\tau}) < 1$,*

there are $K_\infty : \mathbb{C}^n \to \mathbb{C}^n \times \mathbb{C}^n$ and $s_\infty \in \mathbb{R}$ with $1 < s_\infty < \sigma_\omega$ such that

- *(O1) K_∞ is holomorphic on T_{ρ_∞}, primary on T_{ρ_∞} and real on reals;*
- *(O2) $E_{H,K_\infty,s_\infty\omega^*}(\theta) = 0$ for all $\theta \in T_{\rho_\infty}$;*
- *(O3) $H(K_\infty(\theta)) = E_0$ for all $\theta \in T_{\rho_\infty}$ (pointwise);*
- *(O4) $B_{\mu/2}(K_\infty(\theta)) \subset D$ and $\|DK_\infty(\theta)\| \leq \sigma_K - \mu/2$ for $\theta \in T_{\rho_\infty}$;*
- *(O5) $\|K_\infty(\theta) - K(\theta)\| \leq C_2 e/(\gamma^2 \rho^{2\tau})$ for $\theta \in T_{\rho_\infty}$;*
- *(O6) $\|s_\infty \omega^* - s\omega^*\| \leq C_3 e/(\gamma^2 \rho^{2\tau})$.*

Here C_1, C_2, C_3 are chosen after all of $n, \tau, \rho, \rho_\infty, M, \sigma_K, \sigma_B, \sigma_T, \mu, \bar{\gamma}, \sigma_\omega, \nu, \omega^*$ and before $D, H, K, s, \gamma, e, E_0$; hence they may depend on the former, and on none of the latter.

Frozen Lean declaration: `KAM.Iso.isoenergetic_kam`

Source: HL19 Theorem 3.6 and Rem. 3.7, pp. 18–19, in the canonical case $d = n$, $c = h$; proof pp. 45–53. Deviations (all conservative, D-I-G1-UNIFORM, D-I-ENERGY, notes.md): ray data, $\bar{\gamma}$ and the margins are fixed first; the strict inequalities of (3.16) are replaced by $\leq$ (candidate erratum: with $E_c = 0$ the strict bounds read $0 < 0$); the energy conclusion is pointwise.

Remark 1.15 (Reading of I-G1). (a) The frequency of the approximate torus is $\omega = s\omega^*$ on the open ray $\Theta = \{s'\omega^* : 1 < s' < \sigma_\omega\}$. The Diophantine condition is imposed on ω^*, hence (by homogeneity of $k \mapsto k \cdot \omega$) $s'\omega^*$ is nonresonant for every $s' > 0$, with constant $\gamma s'$. (b) ν enters only through (I9) and through the choice of C_1, C_2, C_3; it measures the distance of s to $\partial\{1 < s' < \sigma_\omega\}$ in the metric of $\mathbb{R}^n$ (HL19 H'_5: $\operatorname{dist}(\omega, \partial\Theta) > 0$ made quantitative, $\operatorname{dist}(s\omega^*, \partial\Theta) \geq \nu$). (c) Taking $e = 0$ gives the exact-torus case; then (I13) holds for every C_1, and (O5), (O6) force $K_\infty = K$ on T_{ρ_∞} and $s_\infty = s$. (d) The row-sum bound in (I7)–(I8) is the max-norm operator norm on $\mathbb{C}^n$, $\mathbb{C}^{n+1}$ respectively.

Theorem 1.16 (I-HEAD: all-time action confinement for two degrees of freedom)**.** *Let $B \subset \mathbb{R}^2$ be open, bounded and nonempty; $s_0, r_0 > 0$; h holomorphic on B_{r_0} with $\operatorname{Im} h(I) = 0$ for every real I with $I \in B_{r_0}$ and*

$$\det \mathsf{B}_h(I) \neq 0 \qquad \text{for every real } I \in B_{r_0}.$$

Let $B' \subset \mathbb{R}^2$ be open with $\overline{B'}$ compact and $\overline{B'} \subset B$, and let $\eta > 0$. Then there is $\varepsilon_0 > 0$ such that for every $f : \mathbb{C}^2 \times \mathbb{C}^2 \to \mathbb{C}$ and $\varepsilon \in \mathbb{R}$ with

- *f holomorphic on $T_{s_0} \times B_{r_0}$, $f(\theta + k, I) = f(\theta, I)$ there for $k \in \mathbb{Z}^2$, and $\operatorname{Im} f(\theta, I) = 0$ for real θ and real $I \in B_{r_0}$;*

- $0 \le \varepsilon < \varepsilon_0$ *and* $|f(z)| \le \varepsilon$ *on* $T_{s_0} \times B_{r_0}$,

and every $z_0 = (\theta_0, I_0) \in \mathbb{R}^2 \times \mathbb{R}^2$ *with* $I_0 \in B'$, *there is* $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ *with* $z(0) = z_0$, $\mathrm{Sol}(\mathcal{H}_{h,f}, B, \mathbb{R}, z)$, $\|z(t)_2 - I_0\| < \eta$ *for all* $t \in \mathbb{R}$, *and: for all* $a < 0 < b$ *and all* $y : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ *with* $y(0) = z_0$ *and* $\mathrm{Sol}(\mathcal{H}_{h,f}, B, (a, b), y)$ *one has* $y(t) = z(t)$ *for all* $t \in (a, b)$.

Frozen Lean declaration: `KAM.Iso.all_time_confinement`

Source: ours (D-I-CONFINEMENT): assembled from I-G1, separation in an energy cylinder (Arnold-type barriers; cf. CC03 Theorem 1.1, p. 7, a numerical one-energy precedent with a truncated potential, not used) and ODE continuation.

Remark 1.17 (Reading of I-HEAD). (a) The constant ε_0 depends on B, s_0, r_0, h, B', η only (it may not depend on f, ε, z_0). (b) $\mathsf{B}_h(I)$ is the matrix of Definition 1.7 with $n = 2$; its determinant is the isoenergetic nondegeneracy of HL19 H'_4 at the torus $\theta \mapsto (\theta, I)$: for $H = h$, $K(\theta) = (\theta, I)$, $\omega = \nabla h(I)$ one computes $N_{\cdot b} = (0, e_b)$, $T = D(\nabla h)(I)$, last row $\nabla h(I)^\top$, so $\mathsf{T}_h = \mathsf{B}_h(I)$ exactly (to be proved in part p1, see PLAN.md). (c) The conclusion contains global existence for all real t and uniqueness among every solution on any interval $(a, b) \ni 0$ that remains in $\{I \in B\}$; in particular it contains uniqueness of z among solutions confined to B. (d) The norm $\|\cdot\|$ is the max norm on $\mathbb{R}^2$.

Theorem 1.18 (I-KEPLER: the classical isoenergetic determinant)**.** *For all* $\kappa, \Omega > 0$ *and* $L > 0$*:* $\det \mathsf{K}_{\kappa,\Omega}(L) = 3\kappa\Omega^2/L^4$ *and* $\det \mathsf{K}_{\kappa,\Omega}(L) > 0$.

Frozen Lean declaration: `KAM.Iso.kepler_bordered_det`

Source: Chi05 p. 17 ($\kappa = \Omega = 1$*:* $3/L^4$*); general* κ, Ω *is the same cofactor computation (ours).*

Theorem 1.19 (I-R3BP: near-circular region of the restricted three-body problem)**.** *Let* $a_-, a_+, e_-, e_+ \in \mathbb{R}$ *with* $0 < a_- < a_+$, $0 < e_- < e_+ < 1/4$ *and* $a_+(1 + e_+) < 1$. *Then the set* $\mathcal{N} = \mathcal{N}(a_-, a_+, e_-, e_+) \subset \mathbb{R}^2 \times \mathbb{R}^2$ *is open and nonempty, and there are* $\mu_0, d, R \in \mathbb{R}$ *with* $0 < \mu_0 \le 1/2$, $d > 0$, $R > 0$ *such that for every* $\mu \in [0, \mu_0)$ *and every* $z_0 \in \mathcal{N}$ *there is* $z : \mathbb{R} \to \mathbb{R}^2 \times \mathbb{R}^2$ *with* $z(0) = z_0$, $\mathrm{RSol}_\mu(\mathbb{R}, z)$ *and for every* $t \in \mathbb{R}$

$$\mathrm{rad}\big(q(t) - a_1(\mu)\big) \ge d, \quad \mathrm{rad}\big(q(t) - a_2(\mu)\big) \ge d,$$
$$\mathrm{rad}(q(t)) \le R, \quad \mathrm{rad}(p(t)) \le R, \qquad z(t) = (q(t), p(t)),$$

and: for all $a < 0 < b$ *and* y *with* $y(0) = z_0$ *and* $\mathrm{RSol}_\mu((a, b), y)$, $y(t) = z(t)$ *on* (a, b). *Here* μ_0, d, R *depend on* $a_\pm, e_\pm$ *only, not on* μ *or* z_0.

Frozen Lean declaration: `KAM.Iso.restricted_near_circular`

Source: ours (D-I-PHYSICAL), derived from I-HEAD, I-KEPLER and the Delaunay coordinates of Chi05 §2–3, pp. 8–9, 12–17; not the numerical Victoria theorem of CC03.

Remark 1.20 (Reading of I-R3BP). (a) $\mu = 0$ is included (a massless secondary at $(1, 0)$). (b) The hypothesis $a_+(1+e_+) < 1$ says that the apocentre radius $a(1+e)$ of every orbit of the region is below $1-$margin, so the orbits stay inside the orbit of the secondary; $e_+ < 1/4$ and $e_- > 0$ keep the orbits away from the singularities $e = 0$, $e = 1$ of the Delaunay chart. (c) The region is μ-independent and referenced to $\mu = 0$ (the primary at the origin); the primaries are at $a_1(\mu)$, $a_2(\mu)$ in the dynamics. (d) Collision exclusion for all times is part of the conclusion (distances $\ge d$).